Activity-on-Arrow-Networks

- CPM and PERT methods employ Activity-on-Arrow-Networks (AOAN)
- The concept of events are used in these networks
- In AOAN, Activities are represented as links and the nodes represent starting and finishing events of activities or group of activities

Example Project to draw an AOAN

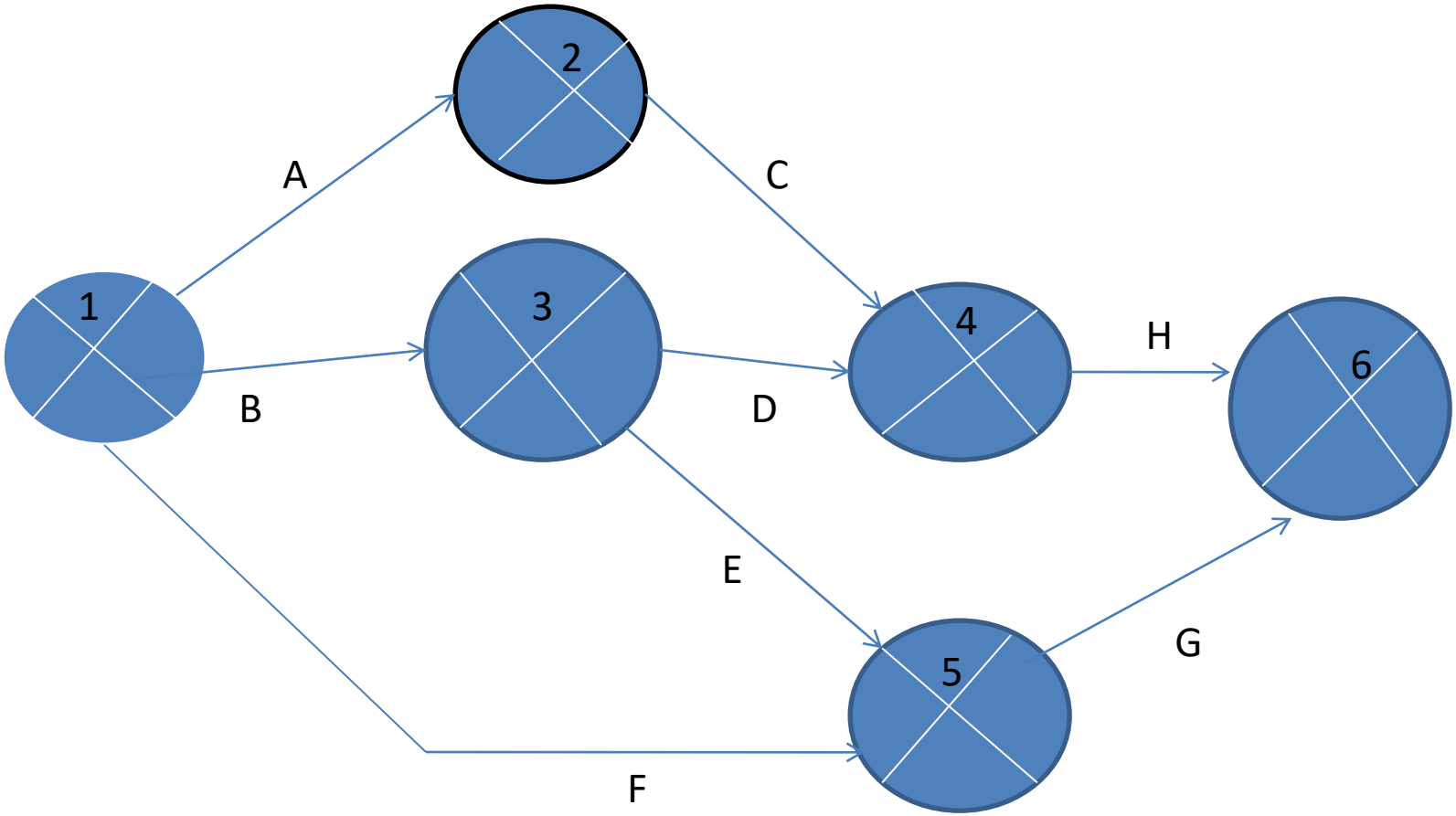
Problem: Consider the activities, duration in weeks and precedents in the Table 1 and draw an activity on arrow network

The AOAN is shown in Figure 1.

Table 1.

Activity	Duration (weeks)	Precedents
A Hardware selection	6	
B System configuration	4	
C Install hardware	3	A
D Data migration	4	B
E Draft office procedures	3	B
F Recruit staff	10	
G User training	3	E, F
H Install and test system	2	C, D

Figure 1. An Activity on Arrow Network



Interpretation of Figure 1.

- The network shows the order in which activities are to be carried out
- 1 is the starting event for activities A and B
- 2 and 3 are ending events for A and B respectively
- C can start after completion of A
- D can start after completion of B
- H can start after completion of C and D

AOAN rules and conventions

- AOAN is required to have only one start node and only one end node
- A link has duration but nodes have no duration
- The start node or the source node is the event of the project becoming ready to start
- The end node or the sink node is the event of the project becoming completed
- Intermediate nodes represent the start or end of an activity

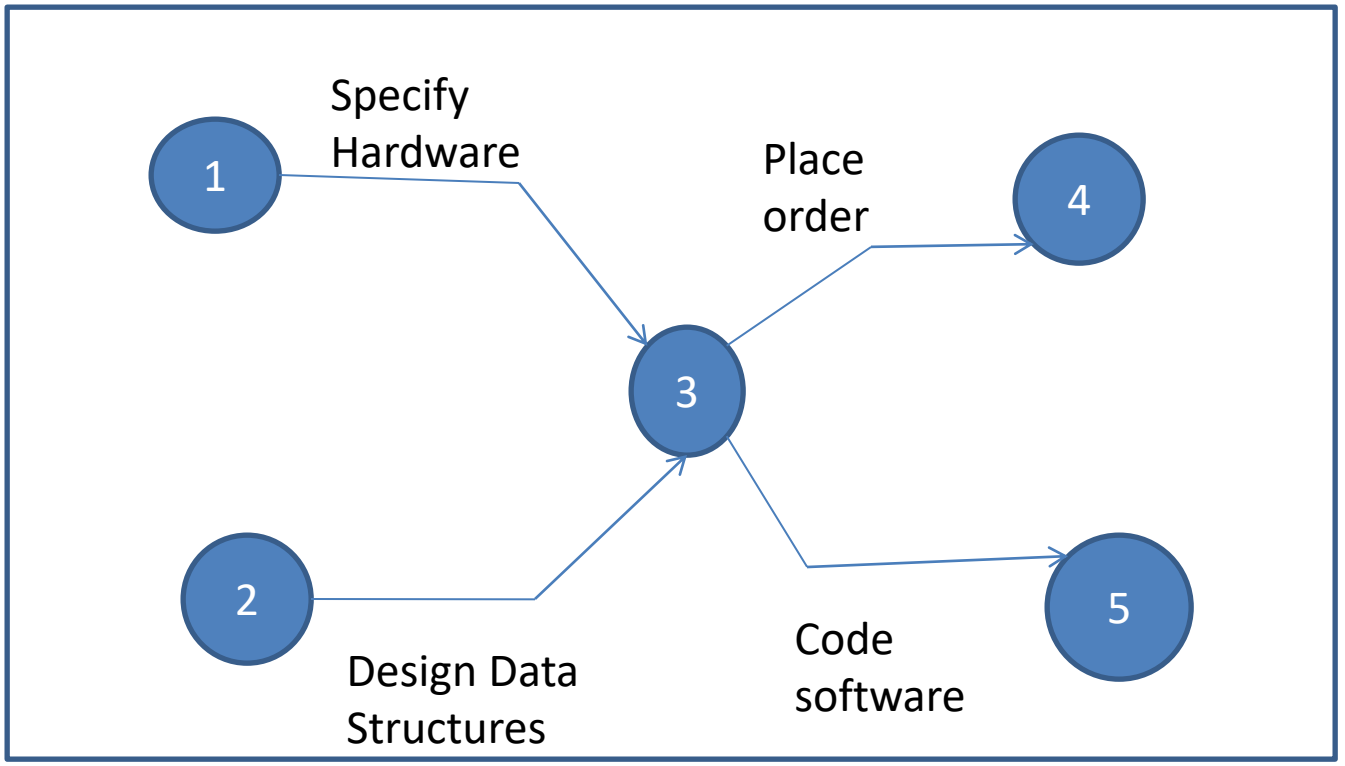
AOAN rules and conventions contd..

- Time moves from left to right
- Nodes are numbered sequentially; nodes at the arrow end of an activity (head nodes) have a higher number than the nodes at the non-arrow end of an activity (tail nodes)
- A network may not contain loops as loops are either an error of logic or an impossible sequence; it must be resolved by mentioning the appropriate number of iterations
- A network may not contain dangles, as it would suggest two completion points for the project

Using Dummy Activities

- When two paths within a network have a common event, a logical error as shown in Figure 2 might occur

Figure 2



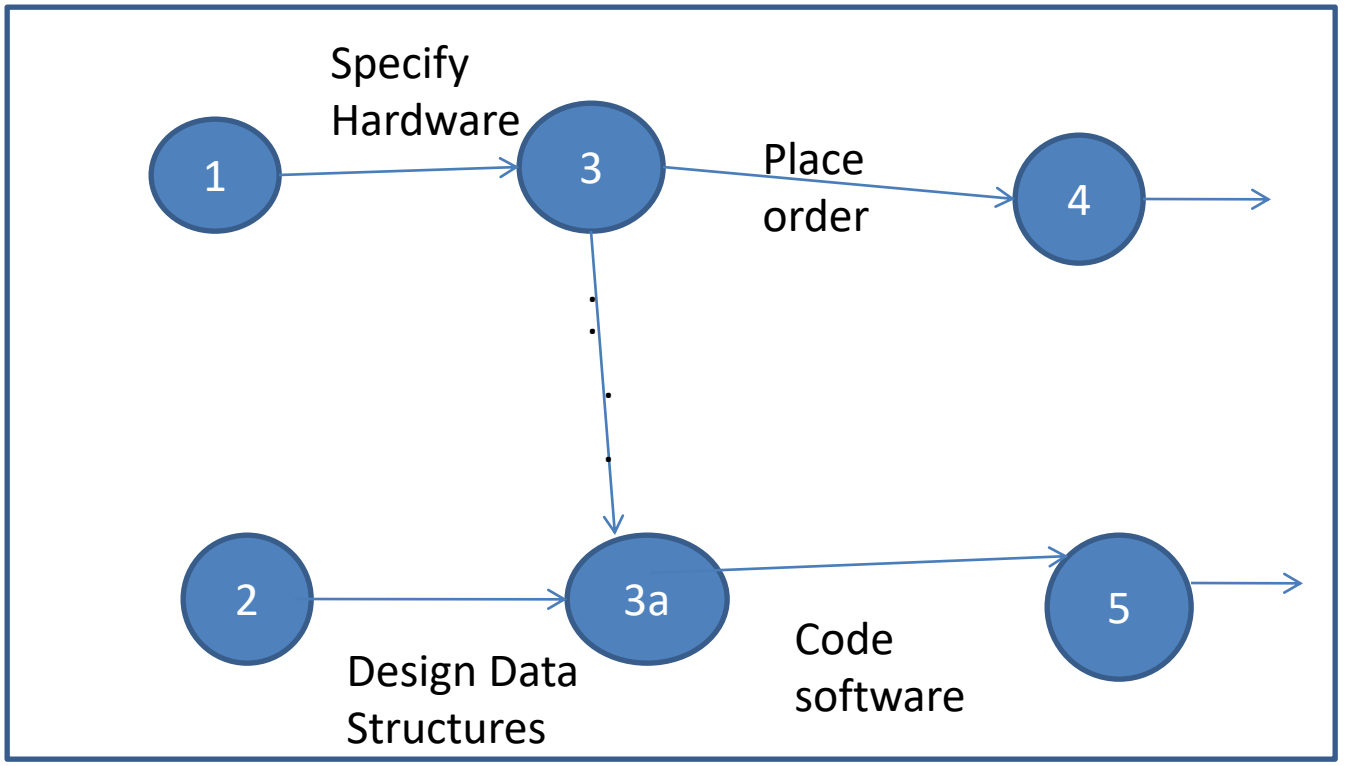
Using Dummy Activities contd..

- In Figure 2 node 3 is the common event for nodes 1 and 2; as a result *specify hardware* and *design data structures* are to be completed before place order code software activities start
- In actual situation only *specify hardware* is to be finished before place order, and *specify hardware* and *design data structures* are to be completed before *code software*

Using Dummy Activities

- The error could be resolved by introducing a dummy activity with zero duration and no requirement for resources as shown in Figure 3. Dummy activity is shown with a dotted line

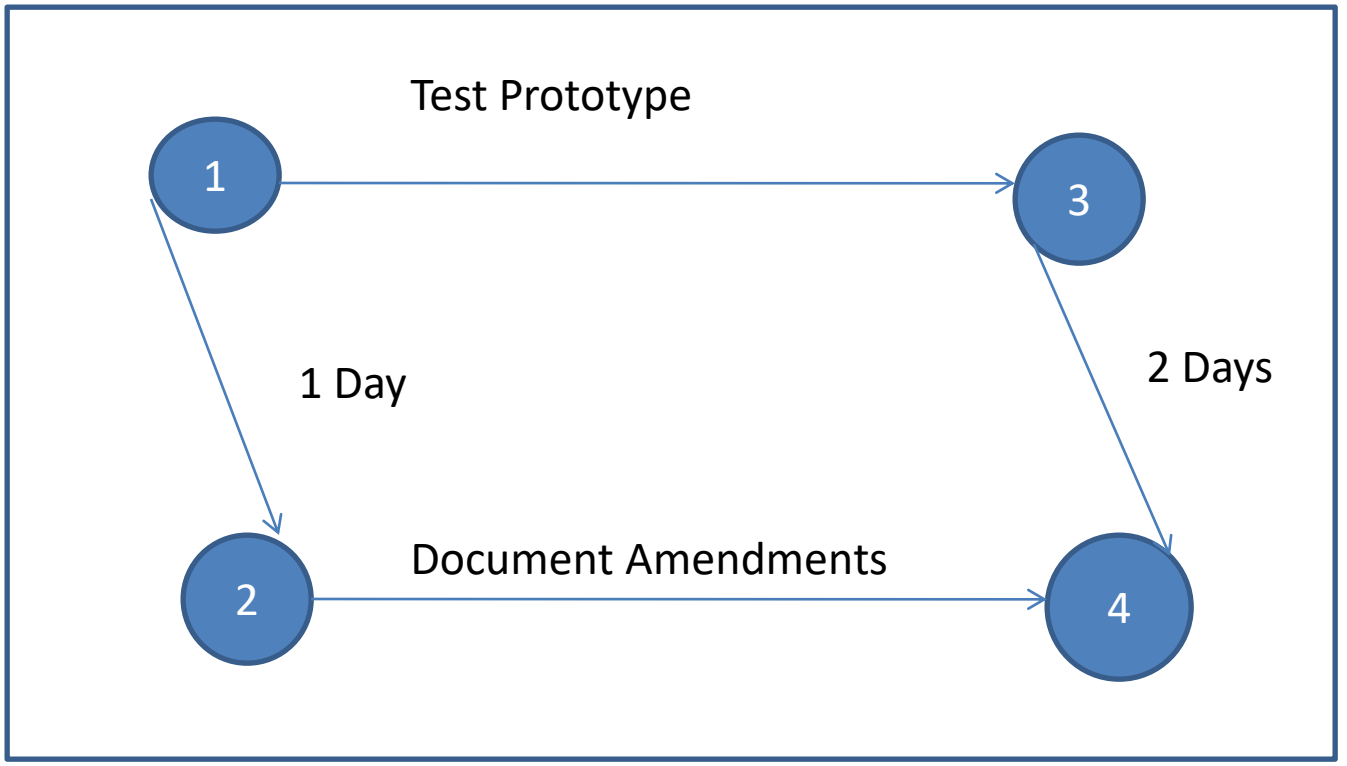
Figure 3.



Representing Lagged Parallel Activities

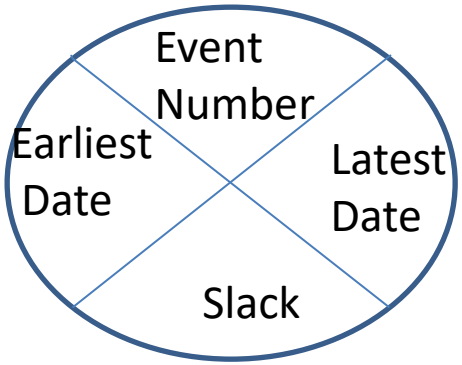
- Lagged Parallel activities are those activities that can be processed in parallel but a stage in one activity must be completed before the other activity can proceed
- Ladder technique is used and each stage is shown as a separate activity in the network as shown in Figure 4.

Figure 4.



Activity Labelling

- Activity based information is generally presented in a different table
- A number of different conventions may be used for representing information on AOAN
- One common convention to label nodes is to divide the node into 4 quadrants and to show event number, the latest and the earliest dates by which the event should occur and the event slack



AOAN Analysis

- The analysis is similar to activity on node networks except the emphasis is on events rather than activity start and completion times
- **The Forward Pass**
- The forward pass is carried out to calculate the earliest date on which each event may be achieved and the earliest dates on which each activity may be started and completed
- The earliest date for an event is the earliest date by which all activities upon which it depends can be completed
- Based on Table1. and Figure 1., the calculation proceeds as follows

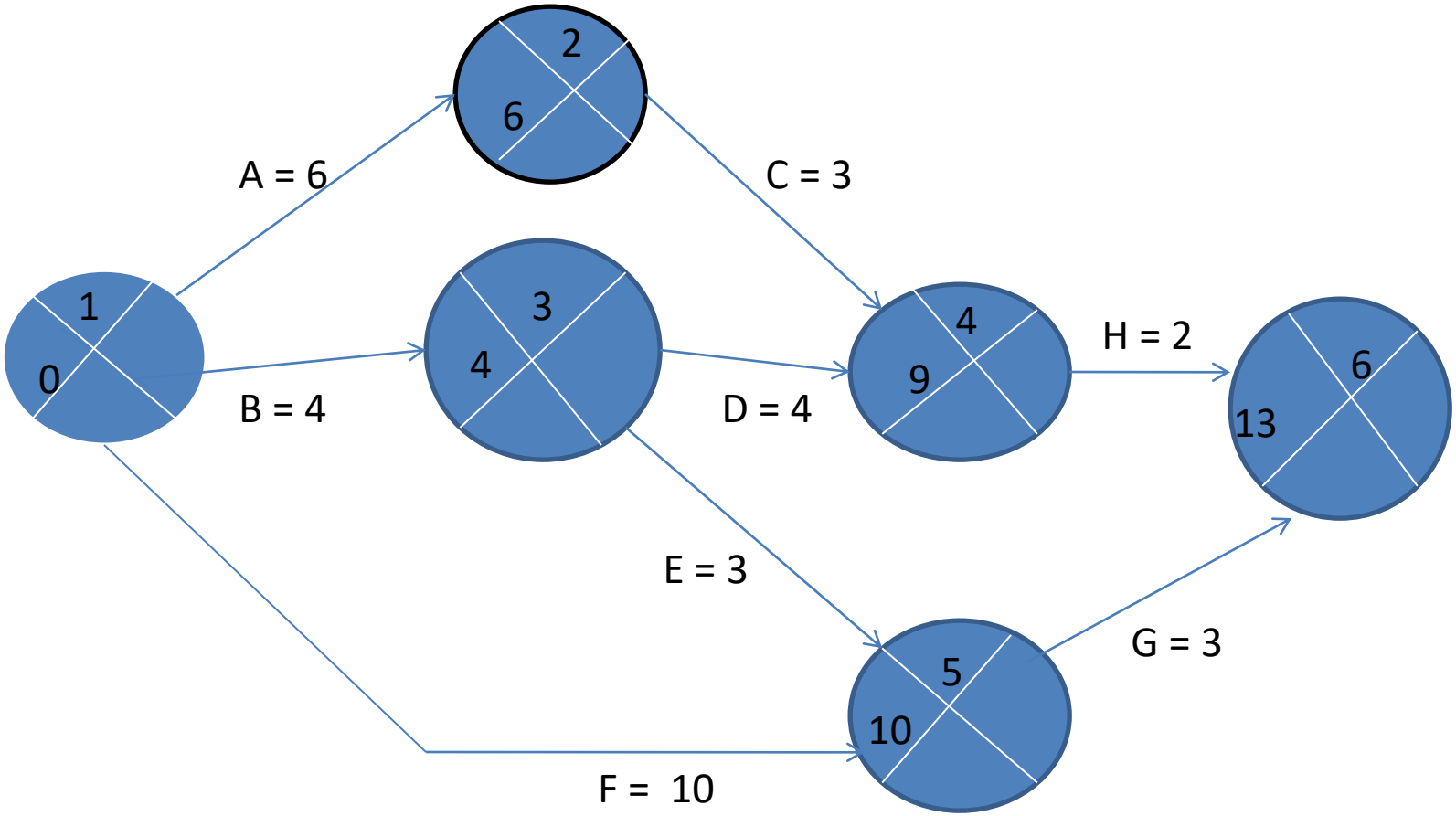
Computations for Forward Pass

- Activities A, B and F may start immediately, so the earliest date for event 1 is zero and the earliest start date for these three activities is also zero
- Activity A will take 6 weeks, so the earliest it can finish is week 6, hence the earliest date for event 2 is week 6
- Activity B will take 4 weeks, so the earliest it can finish and the earliest for event 3 to be achieved is week 4

Computations for Forward Pass contd..

- Activity F will take 10 weeks so the earliest it can finish is week 10; the earliest date for event 5 depends on activity E
- Activity E can start as early as week 4 and since it takes 3 weeks, E will be finished at the earliest at the end of week 7
- Thus we observe that the earliest date for the completion of the project (event 6) is at the end of week 13 – the latter of 11 (the earliest finish of H) and 13 (the earliest finish for G)
- The network after the forward pass is shown in Figure 5.

Figure 5. A CPM network after the Forward Pass



Backward Pass

- The second stage is to perform backward pass to calculate the latest date at which each event may be achieved,

  each event started, and finished,

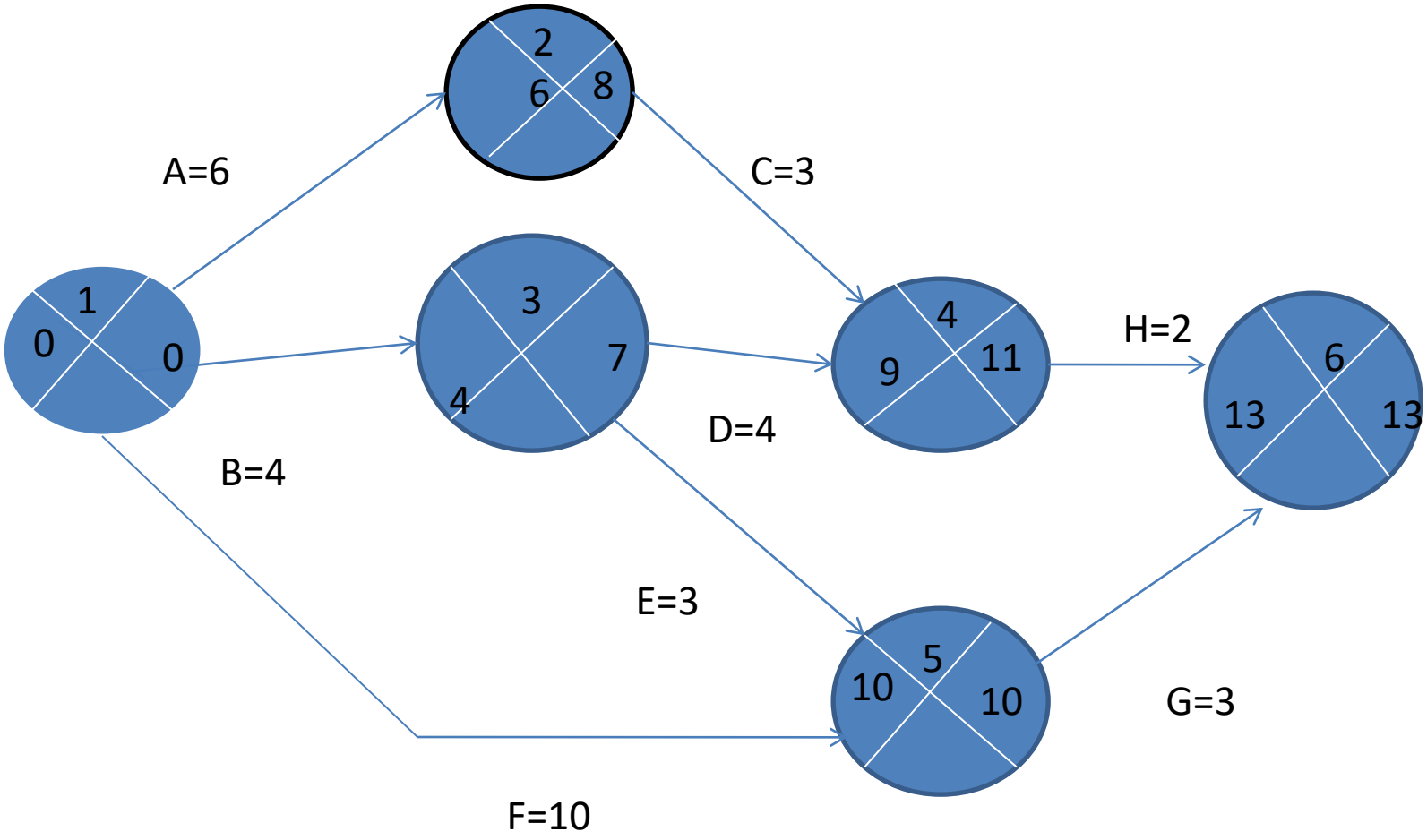
  without delaying the end date of the project

The latest date for an event is the latest date by which all immediately following activities must be started for the project to be completed on time

The latest finish date for the project is same as earliest finish date of the project

The network after the backward pass is shown in Figure 6.

Figure 6. A CPM Network after the Backward Pass



Interpretation of Figure 6.

- The latest completion date of the event 6 i.e. the completion of the project is 13
- As the durations for H and G are 2 & 3 respectively, the latest start dates for events 4 and 5 are 11 (=13-2) & 10 (=13-3) respectively
- In a similar way, the latest start and completion dates for other events are obtained
- The latest start date of the event 1 is zero as the latest start dates of events 2, 3, and 4 are 6, 4, and 10 which are the durations of the activities A, B and F
- If the project does not start on time, it can not finish on time

Identifying Critical Path

- We use slack to identify the critical path
- Slack is the difference between the earliest date and the latest date for an event

  Slack is a measure of how late an event may be without affecting the end date of the project
- The critical path is the path joining all nodes with a zero slack
- The critical path is the longest path through the network
- The critical path is shown in Figure 7.The activity table after forward and backward passes is shown in Table 2.

Figure 7. The Critical Path in AOAN

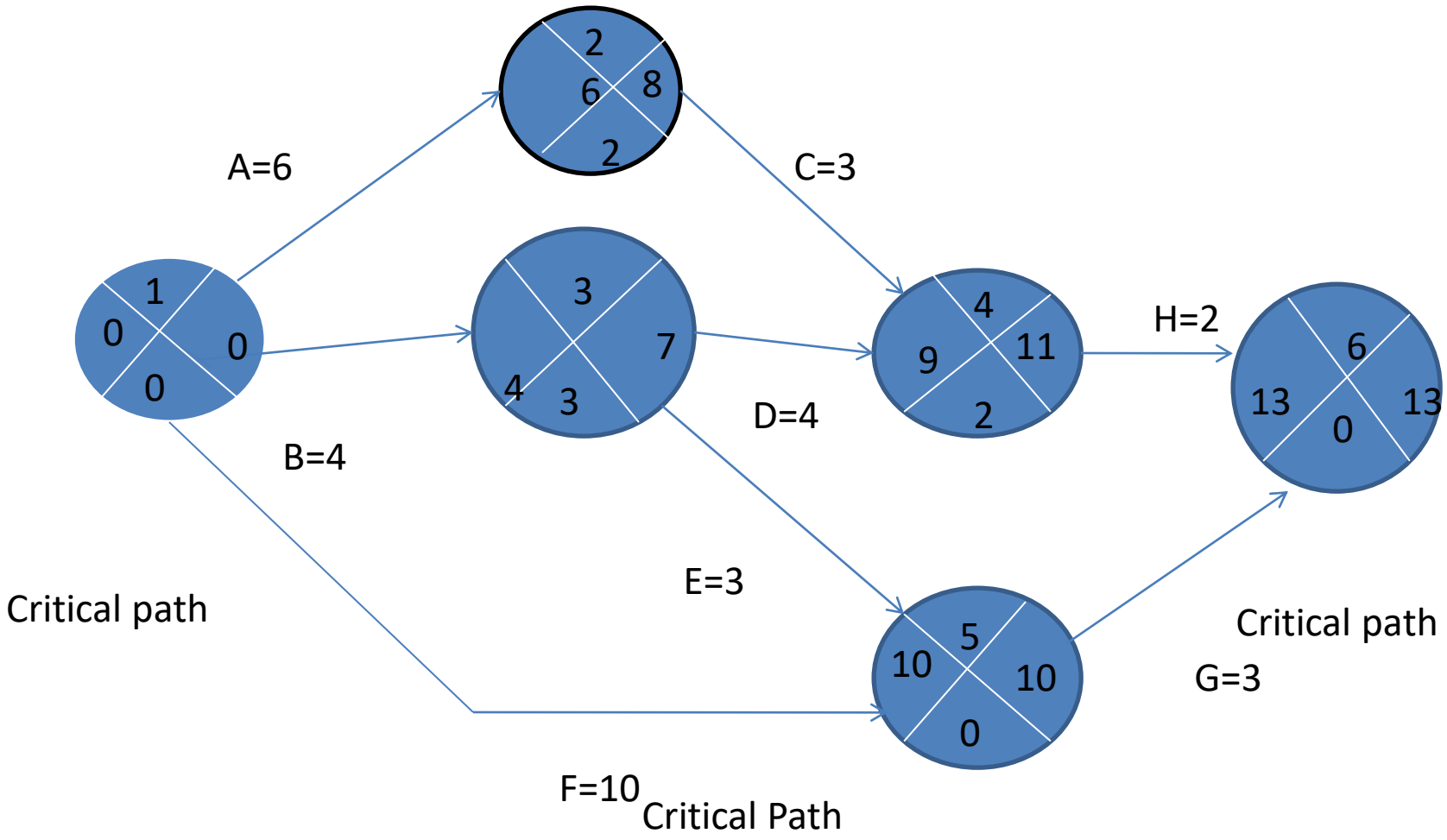


Table 2. Activity Table

Activity	Duration (weeks)	Earliest start date	Earliest Finish Date	Latest Finish Date
A Hardware selection	6	0	6	8
B System configuration	4	0	4	7
C Install hardware	3	6	9	11
D Data migration	4	4	8	11
E Draft office procedures	3	4	7	10
F Recruit staff	10	0	10	10
G User training	3	10	13	13
H Install and test system	2	9	11	13

PERT Technique

- Project Evaluation and Review Technique (PERT) was developed to take account of the uncertainty in estimates of task duration
- It was developed in an environment of high-risk, expensive and state of the art projects
- First Developed for Fleet Ballistic Missile program
- Developed at the same time as CPM
- AOANs are used but the method can be equally well supported by activity on node diagrams

PERT Technique contd..

- PERT is similar to CPM but instead of a single estimate for the duration of each task,

PERT requires three estimates

1. **Most likely time:** Expected time for the task under normal circumstances; denoted by **m**
2. **Optimistic time**: The shortest time in which the activity is expected to complete; denoted by **a**
3. **Pessimistic time:** The worst possible time; denoted by **b**

PERT time estimates and Expected Duration

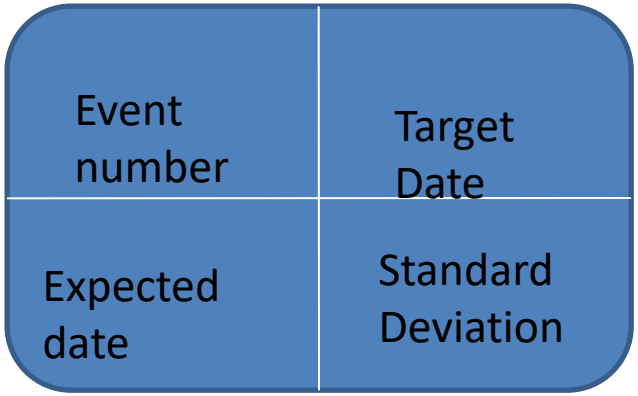
- These three estimates are combined to form a single expected duration t_e using the formula

$t_e = (a + 4m + b) / 6$

The expected durations are used to carry out a forward pass through a network using the same method as the CPM technique

PERT network after the Forward Pass is shown in Figure 8.

The event node is labeled as follows



s standard deviation

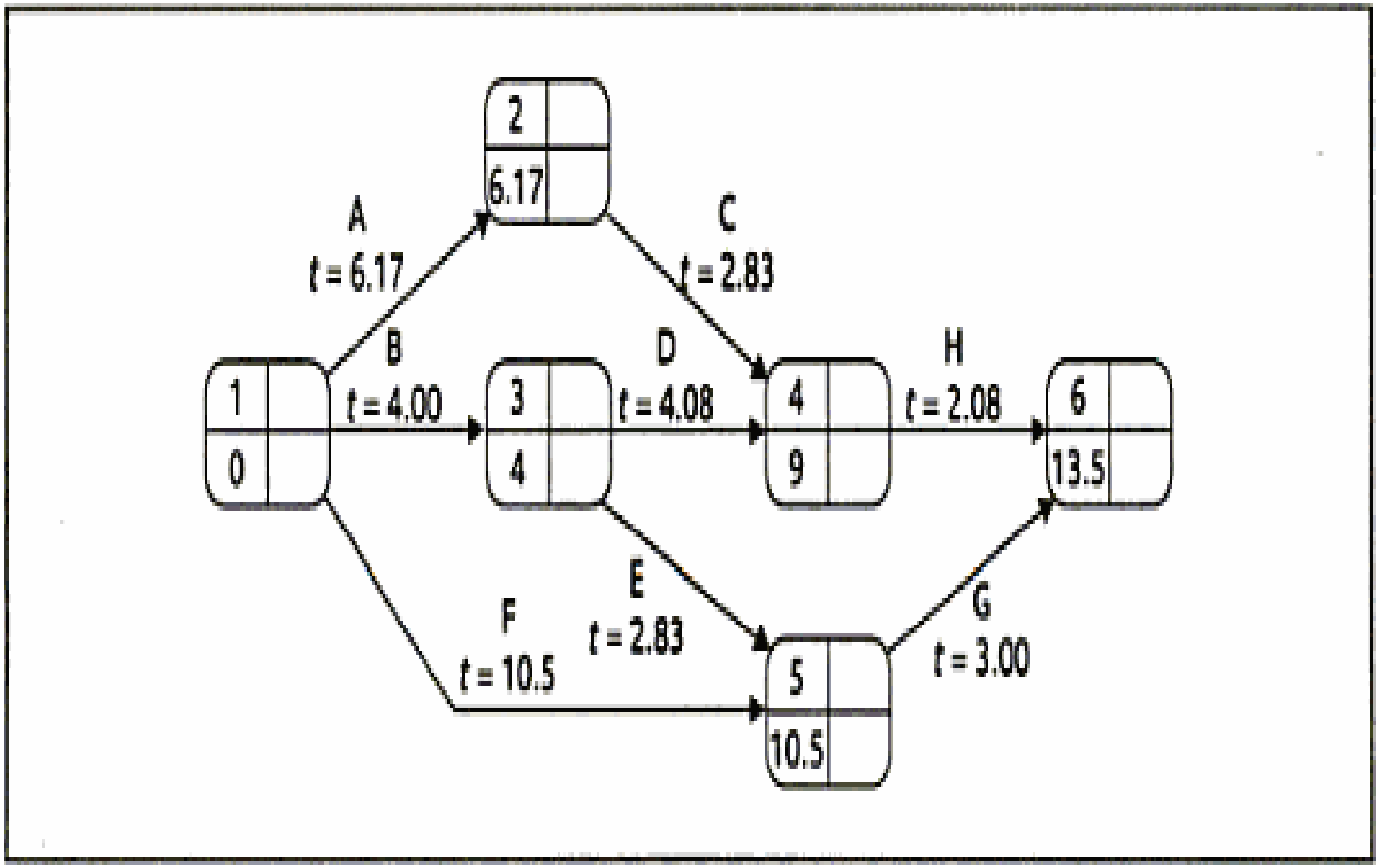
$s = (b-a)/6$

Table 3. PERT Activity Time Estimates for the project in Table 1.

Original activity durations are used as the most likely times, m

Activity	Duration (weeks) m Most Likely	Optimistic (a)	Pessimistic (b)		
A Hardware selection	6	5	8		
B System configuration	4	3	5		
C Install hardware	3	2	3		
D Data migration	4	3.5	5		
E Draft office procedures	3	1	4		
F Recruit staff	10	8	15		
G User training	3	2	4		
H Install and test system	2	2	2.5		

Figure 8. PERT network after the Forward Pass



Interpretation of Figure 8.

- The expected duration t_e for each activity is computed based on a, m, and b using

  $t_e = (a + 4m + b) / 6$

For example, for activity A, $t_e = (5+24+8)/6=6.17$

For activity B, $t_e = (3+16+5)/6 = 4.0$

For activity C, $t_e = (2 +12+3)/6 = 2.83$

Similarly for other activities expected duration is computed

For activity A, the expected duration is 6.17, and for activity C, expected duration is 2.83, for event 4, expected time is maximum of (6.17 + 2.83 = 9) and (4+ 4.08 = 8.08) is 9

Interpretation of Figure 8.

- For event 5, expected time is maximum of 10.5 and 6.83 (= 4 + 2.83)
- For event 6, expected time is maximum of 13.5 and 11.08 (= 9+2.08)
- Thus the expected completion time for the project is 13.5

Activity Standard Deviations

- Standard deviation is a quantitative measure of the degree of uncertainty
- Standard deviation is computed as

$s = (b – a / 6$

The activity expected durations and standard deviations are shown in Table 4.

Table 4. Activity Expected Times and standard deviations

Activity	Activity durations (weeks)				
	Optimistic (a)	Most likely (m)	Pessimistic (b)	Expected (t_e)	Standard deviation (s)
A	5	6	8	6.17	0.50
B	3	4	5	4.00	0.33
C	2	3	3	2.83	0.17
D	3.5	4	5	4.08	0.25
E	1	3	4	2.83	0.50
F	8	10	15	10.50	1.17
G	2	3	4	3.00	0.33
H	2	2	2.5	2.08	0.08

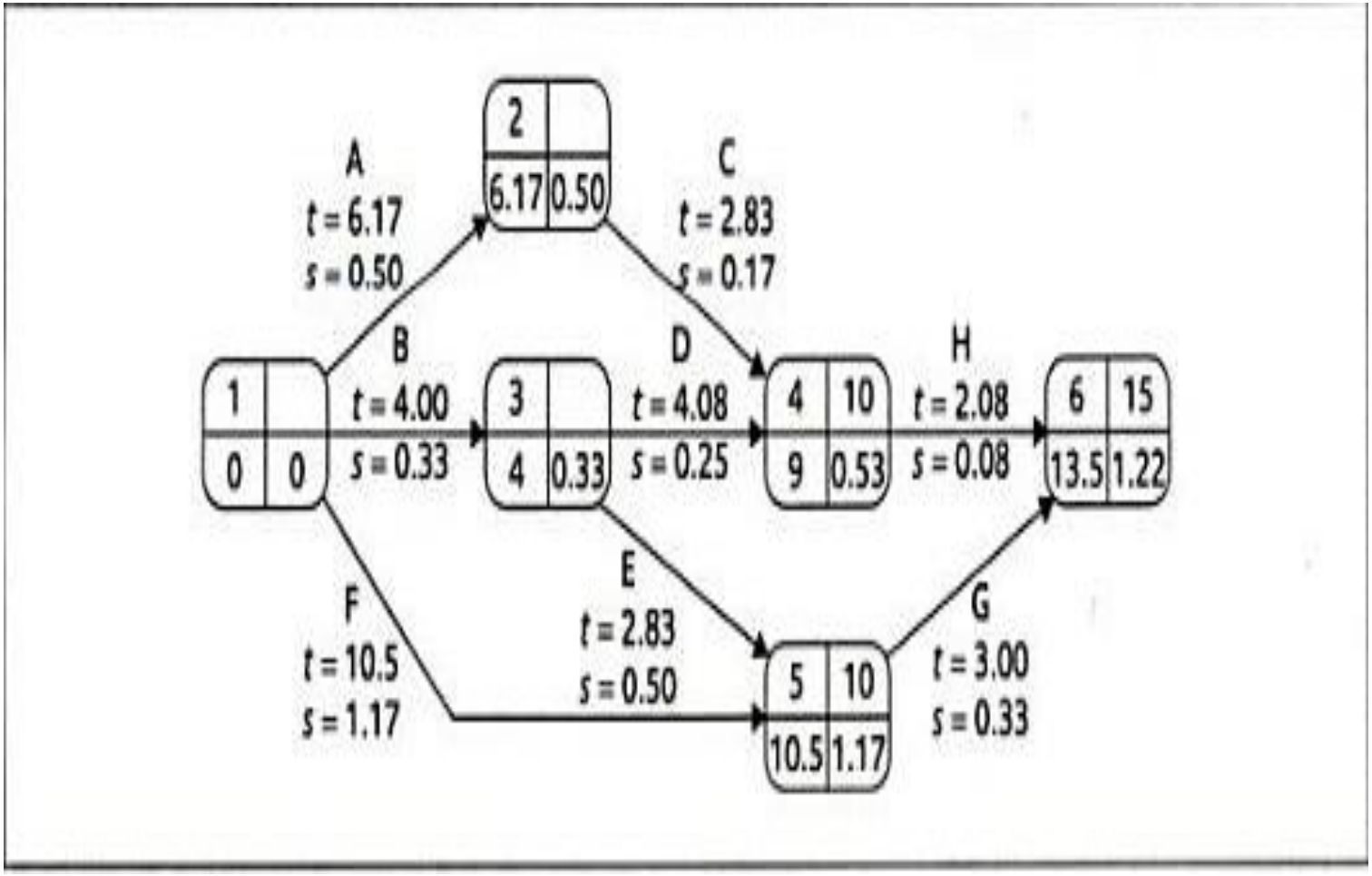
Meeting Targets

- PERT provides a method for estimating the probability of meeting the target dates
- Suppose there are three targets to be met

1. The project must be completed within 15 weeks (expected time is only 13.5 weeks)
2. Activity C must be completed by week 10
3. Event 5 (delivery of intermediate products to customer) must take place by week 10

These target dates and computed standard deviations of Events are shown in Figure 9.

Figure 9. PERT network with Target Dates and Standard Deviations



Standard Deviation of Project Event

- From activity standard deviations, event standard deviations are computed as follows
- Standard deviation (s) for event 3 depends on activity B, so s =0.33 (from Table 4.)
- For event 5, there are two paths: B + E or F
- s for path B + E is $\text{sqrt}(0.33^2 + 0.50^2) = 0.6$
- s for path F is 1.17
- Hence s for event 5 is maximum of s for the two paths =1.17
- Other s are computed in a similar way

Calculating the z value

- z value signifies the probability of meeting the targets
- z is the number of standard deviations between the node's expected and target dates
- z value is computed for each node that has a target date
- z is computed as

  $z = (T - t_e) / s$ where T is the target date and t_e is the expected date

z for event 4 is $(10 - 9.00)/0.53 = 1.8867$

z value for event 6 (project completion) is 1.23

z values and probabilities

- There are standard tables for getting probability for a given z value
- Graph is plotted based on tables
- For the z value 1.23 (project completion), the probability is approximately 11%
- It means there is a 11% risk of not meeting the target date - week 15
- For the z value 1.89 (event 4), the probability is approximately 5%
- It means there is a 5% risk of not meeting the target date - week 10